INFO3315 Final Report

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Introduction

The problem associated with our choice of our food waste application was the presence of irresponsible food consumption, linked to the 12th UN goal “responsible consumption and production”.

Our motivation behind this arose when we saw the salient statistic of 19% of food being wasted at the consumer level; emphasizing the presence of this issue. As a group we had many stories along the lines of our individual food wastage; through mismanaging food, to forsaking expiry dates among others. This highlighted the opportunity for us to (as some examples) design a solution to target this issue at the household level, providing opportunities for management and recipe-making to optimise available ingredients.

User Groups

Establishing User Groups

The identification of Coles Food Manager’s user groups was essential to ensure that the application meets the needs of its intended audience. To achieve this, the team employed both quantitative and qualitative data-gathering techniques. This approach ensured that diverse user perspectives and behavioural patterns were included and analysed.

An online survey was distributed through university networks and social media platforms. It targets a broad demographic that includes students, working adults and families. The survey collected data on participants’ food purchasing frequency, storage habits, awareness of expiry dates and attitudes toward sustainability. Quantitative results provided insights into how different user segments manage food and identified common causes of waste, such as over-purchasing and lack of date tracking.

To complement the survey, a series of interviews was conducted with a smaller group of participants. These interviews aimed to explore deeper behavioural motivations and opinions toward food waste and technology. Participants described frustrations such as forgetting to consume perishable foods, uncertainty about food freshness and difficulty maintaining consistent shopping habits. The qualitative insights helped to contextualise the survey findings in deeper perspectives.

Identified User Groups

1. General adults

This group includes working adults with demanding schedules who often forget what food items they have at home or when they expire. Their busy lifestyles leave little time for manual tracking or meal planning, leading to unnecessary food waste. The app addresses these challenges with functionalities that support automated expiry reminders and AI freshness recognition. These features can minimise manual input, streamline food management and support users who value convenience and time efficiency.

2. Family managers

The app focuses on parents or caregivers who are responsible for managing household groceries and meals. This user group often struggles with over-purchasing and recipe generation. Thus, the app provides specific functionalities that support dynamic inventory tracking, portion planning and recipe suggestions. These functionalities help users maintain balanced grocery quantities, reduce food waste and generate meal ideas that make efficient use of available ingredients.

3. Students and young renters

This group includes young adults living independently or in shared housing who often face challenges such as irregular meal routines, limited storage space and limited budgets. This app contains functionalities that support quick scanning functions, recipe suggestions base on customised budget and expiry notifications. These features facilitate smarter consumption, reduce spoilage and support sustainable habits without adding complexity to their daily routines.

Key Findings

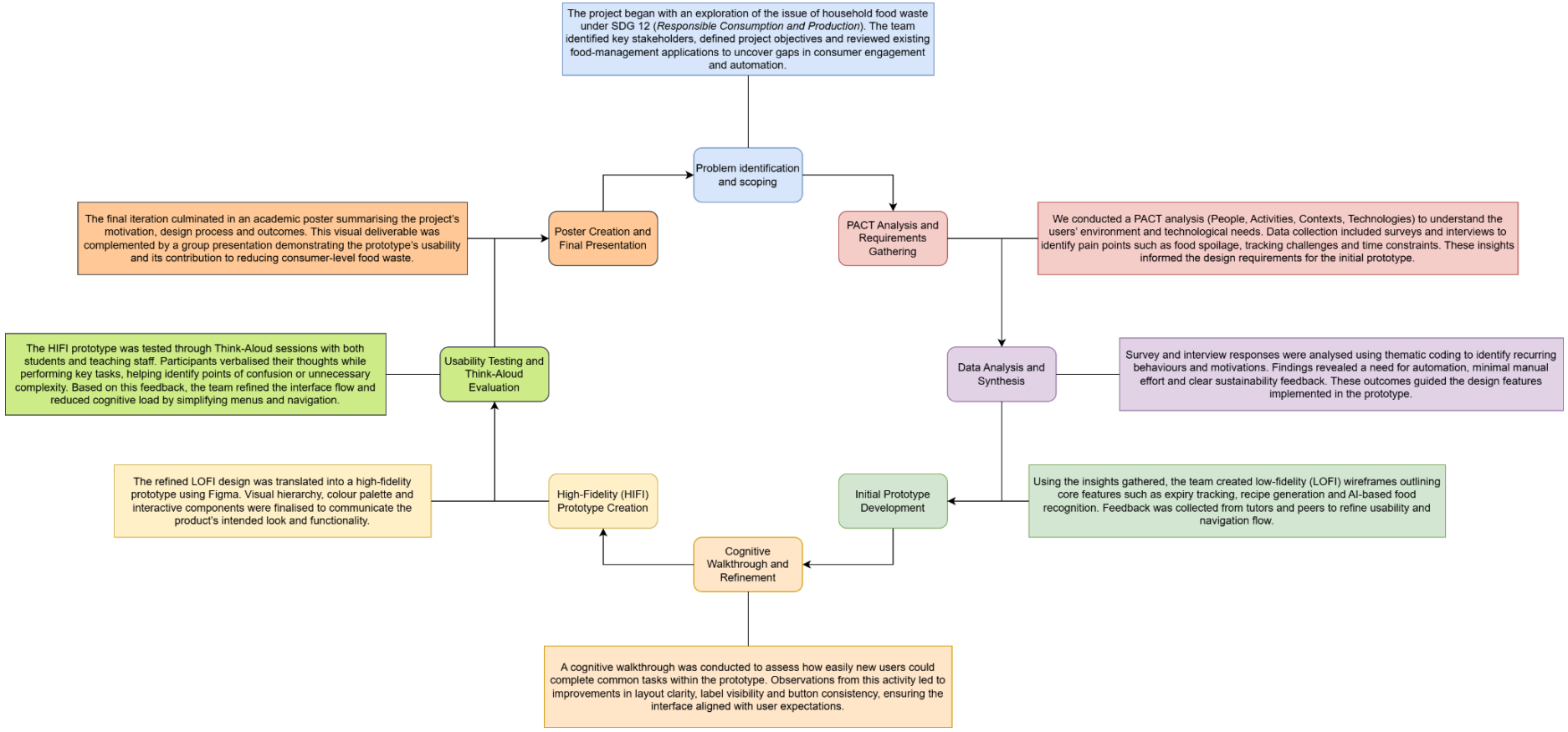
Analysis across all user groups revealed several common issues that inspired the overall application design:

- Users strongly preferred features that reduced manual data entry, such as barcode scanning, AI recognition and automatic reminders.
- Many participants expressed concern about food wastage, however, there are only a few who actively tracked expiry dates, indicating automatic system assistance is essential for everyday food management.
- Users will get motivated by seeing the direct impact of their actions, such as money saved or environmental benefits achieved through waste reduction.

These findings confirmed the importance of designing a system that is intuitive, automated and motivating, encouraging sustainable behaviour through subtle interaction rather than imposing additional effort on the user.

Process Lifecycle

We conducted Problem Identification, PACT analysis, Data Collection, Data Analysis, to inform our first prototype as discussed below. With this prototype we continually iterated it through the below process lifecycle.



https://drive.google.com/file/d/111XLTzlA4Nwpln-0Y8CFSr5NRrj_-D0C/view?usp=sharing (Open draw.io directly from the link for better resolution)

Design Evolution

Each step of the process above from paper prototypes seeked to refine the previous iteration through design principles and guidelines in particular we've learnt in INFO3315, alongside processes learnt such as cognitive walkthrough and usability testing being conducted to improve our design. We go into more detail here about how each was incorporated. We have included information for new users with relevant tools and materials on how to carry out our usability tests so others can execute the plan with similar data.

Initial prototype creation and refinements

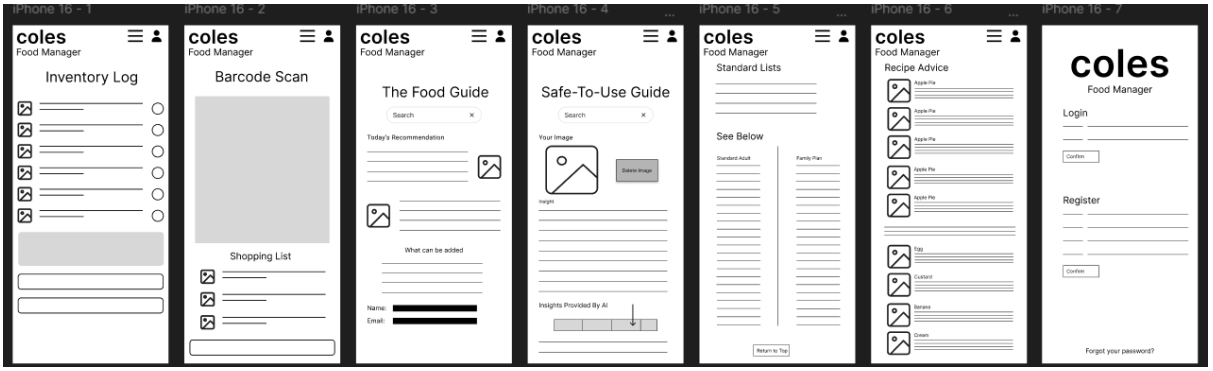
Summary of Process

We developed paper prototypes in order to visualise ideas for our interface and take into account requirements feedback. Based on conceptual models we created we also produced initial storyboards to visualise how we feel our users would interact with it. We also got our tutor to conduct a general usability analysis of these early fragments, confirming the general flow of the paper prototypes are suitable for our target users, and the storyboards are also suitable and the most effective features to target.

Implications

These aspects helped us get our ideas on paper so we could clearly see and understand what we were planning to develop. In addition, feedback from our tutor was invaluable; providing us key tips to influence the direction of our outcome. Taking into account our findings from these paper prototypes, storyboards, tutor feedback and design principles and guidelines we worked as a team to produce our first lo-fi wireframes as seen.

Lo-fi Model One



Cognitive Walkthrough

Summary of Process

Within our team we carried out a cognitive walkthrough on our initial lo-fi wireframe whereby we continually assigned each other different roles, with one person each time acting as the participant, the rest as researchers. Prior to this we had decided on three tasks for users to be able to achieve to

effectively use our application shown below. We use Design Guideline Neilsen's heuristics, and Internal Cognitive framework to assist in informing our evaluations for changes.

Cognitive Walkthrough Summary of Plan (for others' execution)

Process of Cognitive Walkthrough

Cognitive walkthrough is a unique form of usability testing in that it doesn't require external users to test it. The team themselves highlight key tasks, typically three for users to do, and they try to achieve them themselves, writing down their processes in achieving it.

Significance of Cognitive Walkthrough

By writing down those processes, the developers are able to highlight areas where it isn't quite as clear as it should be to achieve tasks and seek to improve them.

Tools and Materials Required

For the cognitive walkthrough, tools and materials required were the laptop / relevant tool itself as a tool so we could write down tasks, and then attempt to carry them out and write down the results. Having two simultaneously made this much easier where one could use the Figma wireframes for the tasks whilst another wrote down the steps to achieve them, and any issues. The key material here was the google doc to notate steps and results.

Tasks

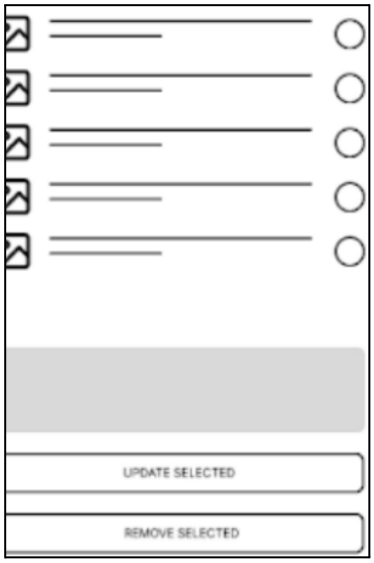
- Deleting items
    - How to delete items from the login page?
    - Significant feature as users need to remove items after consumed
- Barcode scan for updating inventory
    - How to scan new items from the login page?
    - Again important as users need to be able to add items via the barcode scan to get them in the inventory at all
- Safe to use food guide
    - How to navigate to the safe-to-use food guide page?
    - An important feature of the app was to prevent food wastage. The feature to check food is highly useful.

Findings and Implications

- Deleting items
    - Successes
        - In terms of Norman Gulfs of Execution, the users goals seem to generally match well with the system capabilities. The system has clear buttons for selection and a clear delete button.
    - Changes

        - The only problem in terms of execution was not even with the delete items page, but we didn't specify this is the home page when logged in so we've done that in a future iteration. The navbar wasn't incorporated at this point. This encompasses Nielsen's heuristic "Visibility of system status" and as shown below adds clarity.

        - However problems were more with the evaluation of the system by user stage where extra measures such as a notification and double check to alert users before deletion to avoid mistakes was necessary to improve again "Visibility of system status". These functionalities go beyond the scope of a Figma Wireframe, and are something we'd look into in a genuine online solution via programmers.
- Barcode scan for update inventory
    - Successes
        - In terms of the Norman Gulfs of Execution here, the user goal is to accurately scan in new items via a barcode and it appears function-wise this logic appears suitable.
    - Changes
        - In terms of Execution, to improve we feel making clear the image area has a camera with clear camera symbols, and the ability to confirm/ update items via adjust quantity buttons on the right would be worthwhile enhancements. Both were included in the final design as a result, but due to the lofi stage, we didn't bother including the camera feature for the next step yet.



        - For gulfs of evaluation, we feel a notification to say "barcode scan successful" would be useful. In general we think ethically a "Camera consent" notification is imperative here as well. Again notifications go beyond the scope of a Figma outcome but are something we would certainly include if the outcome is brought in as actual software. We did include the barcode successful notification however via some Figma wrangling in our eventual outcome.

- Safe to use guide function
    - Successes
        - The task is overall intuitive with some possibility of minor misunderstanding with the delete image button.
    - Changes
        - Adding an AI statement at the end of AI insights to prevent taking more responsibility which relates to Nielsen heuristic “Help and documentation”, providing important insight to be wary of AI output.

          AI insights not always accurate, consult industry experts for the most accurate information
        - Making adjustments with the delete image button to prevent unnecessary misunderstanding.

Usability Testing with another group

Summary

In the same session we also asked another group to evaluate our current prototype, with an emphasis on areas for improvement.

Usability Testing Summary of Plan (for others’ execution)

Process of Usability Testing

Usability testing is the process of gathering feedback from others to understand and potentially influence our design

Significance of Usability Testing

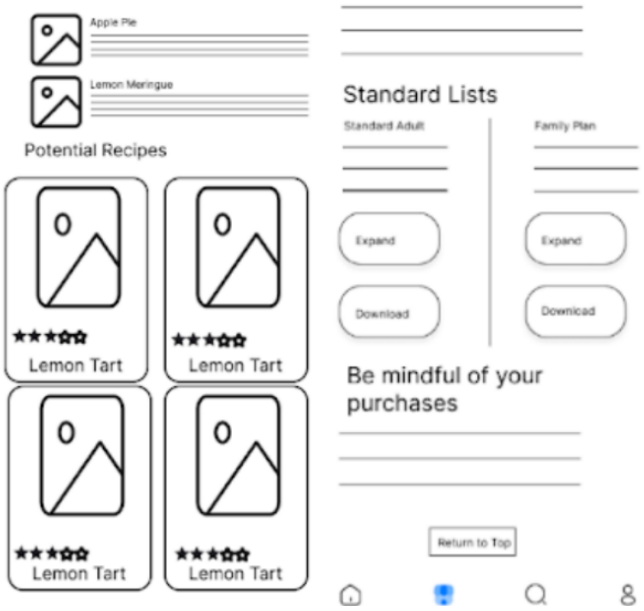
Usability testing helps us understand where our application is thriving and where edits are required, helping us spot things from users without our biases as testers or just in general.

Tools and Materials Required

The key tool is a laptop / relevant tool from the notator who gathers the feedback whilst the consulter discusses with the participant about their views of the app, which will require another laptop / relevant tool to access the prototype. Materials required again will be a google doc / equivalent to access the notations.

Areas of Improvement and Implications

- Some pages are too dense (in particular standard lists and recipe advice) so find solutions if lots of information is needed. Have a pop-up window to prevent over-crowding on the standard lists page. So we made these pages briefer, using more images rather than text for the recipe page, and a pop-up for the shopping tips page.



- Having a navigation bar would improve the UI flow for a phone application tremendously. So we incorporated it.

- Buttons should only become visible upon hovering. Buttons are now just the color of the background with a stroke and text.

- Make the safe-to-use guide have a barcode scanner for consistency. This was included in the first Hi-Fi wireframe version

Team Decisions

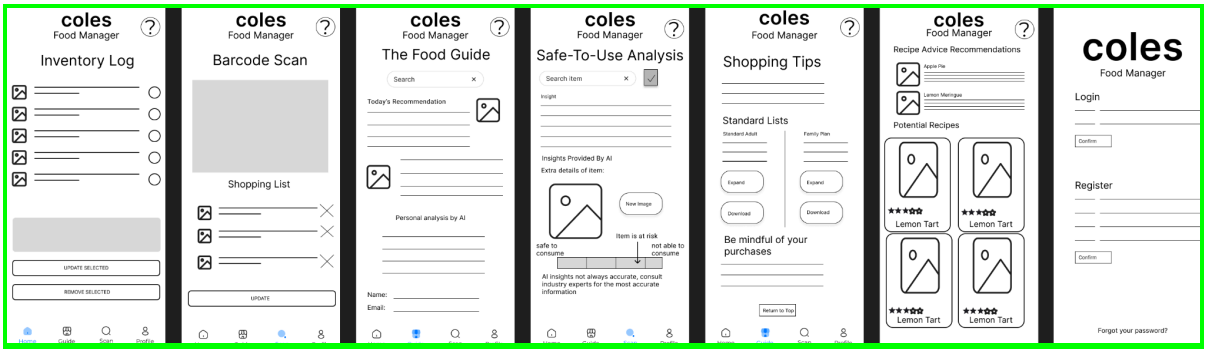
Summary

We then met as a team to discuss any final changes to improve the UI until we complete our Lo-Fi version to be made into Hi-Fi soon.

Findings

- We felt a Login/Register page was necessary so we've added that in.
- We removed the dashboard bar from earlier leaving a gap in the top right. We felt users could struggle with navigation so incorporated an interface metaphor, a question mark, to represent a potential help page.
- Everything's been spaced out more minimalistically, in search of achieving Neilsen heuristic "Aesthetics and minimalist design".
- Added more context for the safe to use analysis
- Made the food guide "what can be added" into more of a "Personal analysis by AI" to be more useful to the user than a blog they can add to; more relevant to our "food waste" topic getting some food advice for a healthy, sustainable diet.

Lo-fi model two (the final lo-fi model)

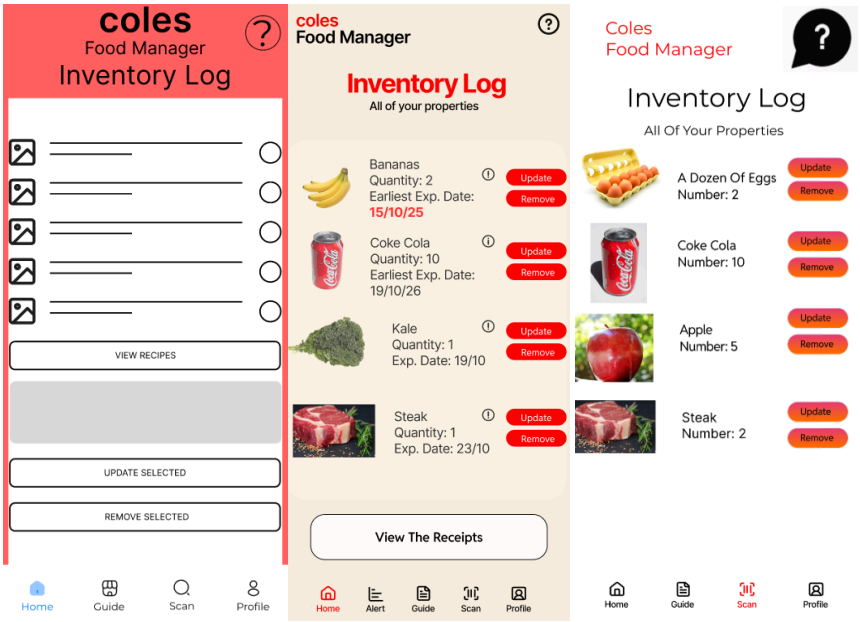


Developing different designs for first hi-fi wireframes

Summary

We then added colours, images and actual textual content in place of the placeholder artifacts. We did this in such a way that encouraged us to be creative. Everyone produced different styles with the same general color scheme style and same fundamental structure from the lo-fi diagram, taking different pages each (some not fully hi-fi yet). We would then compare and go with features we liked of each.

Examples of work produced

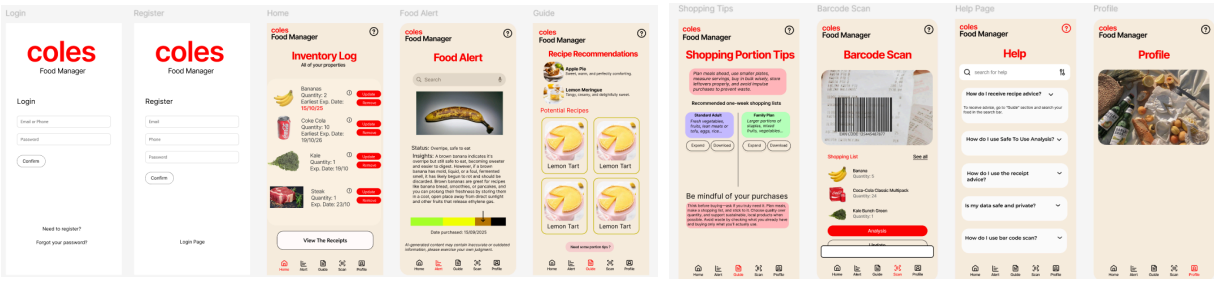


Decision

Ultimately, as a team we decided to go with the middle option as we felt it firmly met "aesthetics and minimalist design" criteria, consistent spacing, and useful functionality we didn't consider which were:

- Information tabs for more details on foods and expiry dates
- A profile page
- Actually including the profile page as we forgot about this.

Hi-fi wireframe one



After we do the LoFi to HiFi Prototyping iteration, we apply the think aloud testing in week 9 tutorial. Our group designed several core features for this hi-fi model, and asked a random student in the lab to work with us.

Think Aloud

Summary

Our group carried out think aloud testing in the tutorial to gauge whether users were able to carry out crucial tasks for our proposed application. After this think-aloud testing we were able to identify and in turn fix many flawed designs, further improving our design.

Think Aloud Summary of Plan (for others' execution)

Process of Think Aloud testing

Think aloud testing involves getting users to speak aloud whilst they seek to achieve tasks of the prototype, three being a recommended amount. Their spoken thoughts along with successes and failures in terms of navigation are notated to spot positives and negatives of the prototype to highlight areas to change.

Significance of Think Aloud testing

Many flawed designs result from the designer's negligence. At this stage, a tester who has never encountered our design model is particularly crucial. Think aloud testing helped us uncover errors that we, as developers, assume are correct but are actually flawed.

Tools and Materials Required

For the think-aloud testing, tools and materials required were the laptop / relevant tool itself so our participant could access the app to carry out tasks, and of course figma. Materials required were someone's separate laptop / relevant tool in order to notate feedback. We separated roles so these roles of session-runner, helpers, and notator could run the sessions efficiently. People from our class were

required as participants, and we ensured we gathered 3 different perspectives to highlight various aspects of our prototype.

Implications

In the think aloud testing session, we required testers to complete the specified functions including sign up, view inventory log, add/delete new items into the inventory, search user's apple in the "Food Alert" page, scan barcode to load receipt for items in the inventory, view standard adult list, and get help. The tester pointed at the model to explain its path and provided our team with some updated suggestions. Tools needed

1. The testers claimed the dashboard page can be clearly identified and it is clear to use the bar code scan to add items in the list but the sliding area is not clearly defined.

   Actions: Using a color with high contrast and the lower half of the last item is covered by the layer above it to indicate that this section is slidable.

2. The testers think it's an intuitive process to scan an apple in the "Food Alert" page to check whether the apple is able to be consumed. At the same time, the tester also warned us about the risk of confusing the two pages between "Food Alert" and "Barcode Scan"

   Action: Add a camera background in the back layer of the picture in the "Food Alert" page to indicate that this is a page which needs to use the camera.

3. The testers said it's clear and intuitive to scan the receipt from the Coles to extract the purchased items.

   Action: We add a label to show the photo is successfully taken on the camera interface.

4. The testers cannot get the actual meaning of "standard adult list" and failed in this task. One test claimed that he thought he needed to go to the "Food Alert" page and this task is not pretty much intuitive and not clear enough

   Action: We add a new button in the bottom of the recipe page called "Need some portion tips ?" to guide users to click through to the page providing purchase quantity recommendations.

5. The testers cannot find the connection between the question mark and the help page, and he said he is not sure about the pathway of this feature.

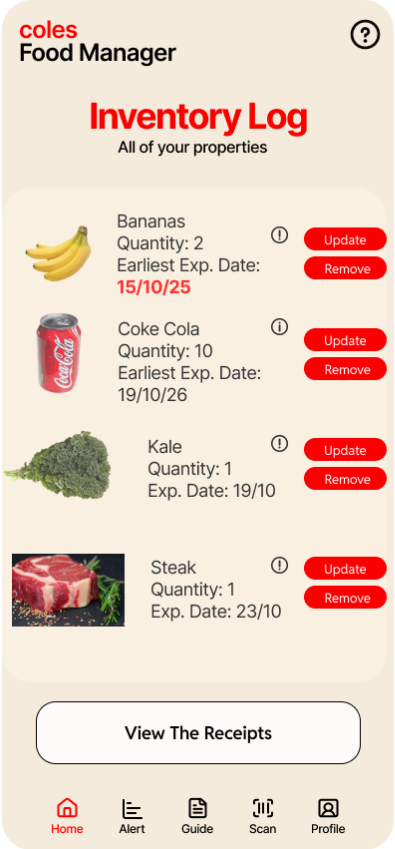
   Action: Add more tips in the "Help" page to indicate the usage of this one.

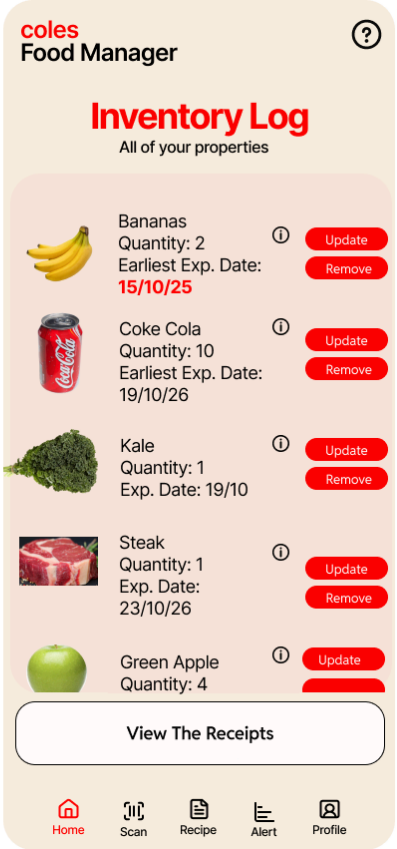
6. The testers find it's hard to understand the logic among the pages for signup, login the the profile page.

   Action: Make the profile page much more detailed with avatar, name, phone number and email address (all invalid and fake in this model).

Following the insights gathered during the initial think-aloud testing session, we have implemented several strategic changes to our hi-fi prototypes. These revisions focus on enhancing clarity, usability, and aesthetic appeal to deliver a more intuitive user experience.

Improved Scrollability and Inventory Management (suggestion 1 from testers):
To clearly communicate which areas of the interface are slidable or contain additional content, we introduced a high-contrast overlay background (light red/pink) on the relevant section. This visual cue serves as an indicator, making the interaction more discoverable. Here, we added an extra item (Green Apple) to the visible list. By increasing the number of displayed items, we reinforce the visual suggestion that the list extends beyond the screen boundaries, confirming the section is scrollable.





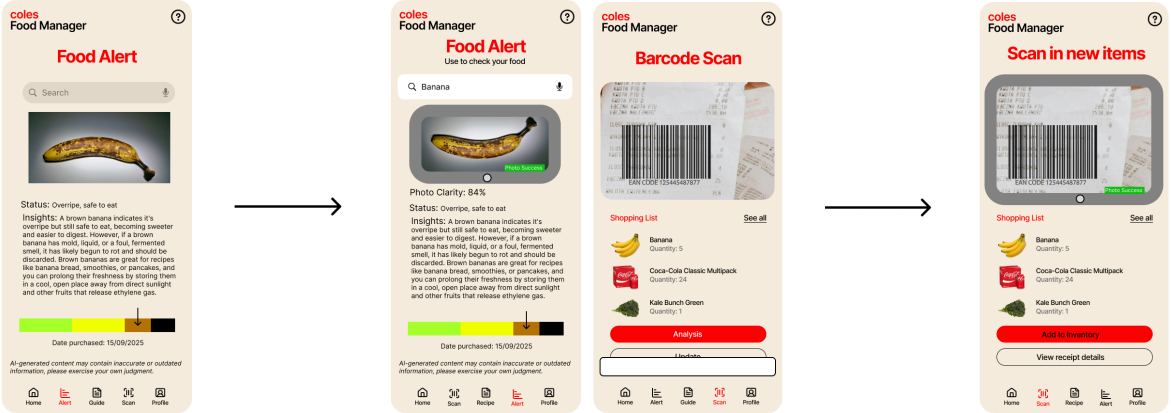
Enhanced Camera and Scanning Feedback (suggestion 2 and 3 from testers):
Based on user feedback, we focused on providing clearer visual feedback during camera interactions:

- **Camera Visuals**: We introduced a clear gray outline and a dedicated camera button icon to represent the live camera view, making the function immediately recognizable. This adheres to the "Recognition over Recall" usability heuristic.

- **Action Feedback**: A small "Photo Success" pop-up label was added to instantly confirm that an action (taking a photo or scanning) has been completed successfully. This instant feedback reduces user ambiguity.

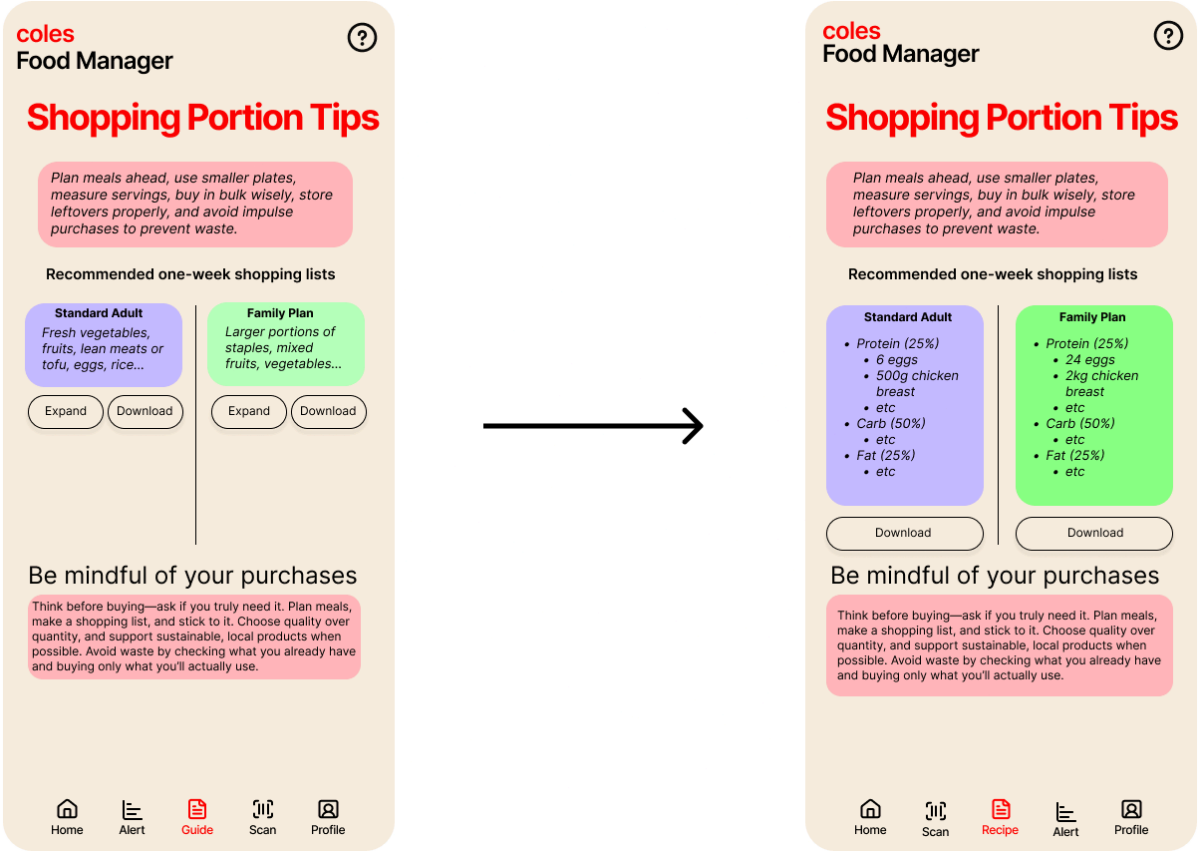
- **Photo Clarity Metric**: For quality control, we included a "Photo Clarity" percentage. This quantitative metric empowers the user to adjust the angle or lighting for a better image, ensuring higher data quality for the underlying object detection/prediction model.

- **Search Bar Prominence**: The search bar was made more noticeable through adjustments to its size and color, improving its discoverability and scannability on the page.



Improve 'Shopping Portion Tips' page

We streamlined the content view. The "Expand" button was deemed redundant and removed, following the Fitts' Law principle by reducing steps. Users can now directly click on the condensed list display to view the full details, a more direct and efficient interaction. Furthermore, we utilized white space more efficiently to display more items on the screen without excessive visual clutter.



Improve 'Recipe' page

The Recipe page underwent significant changes to better demonstrate its functionality and provide a richer, more diverse experience:

- **Visual Structure**: A border was added to each recipe card to improve the Gestalt Principle of Common Region, clearly defining separation and improving content organization.

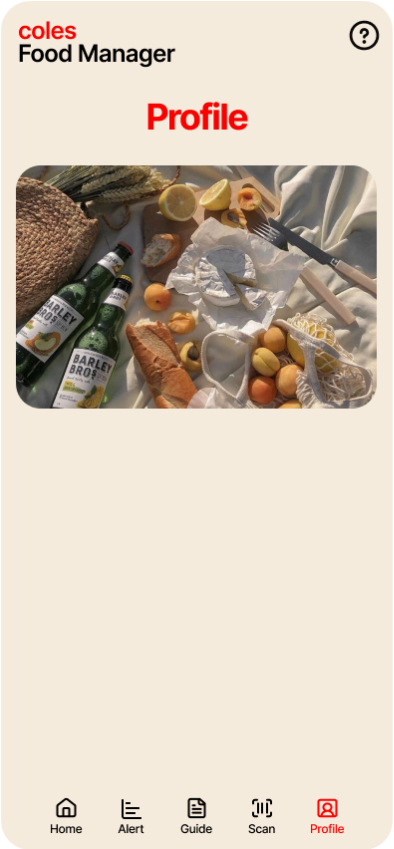
- **Recipe Complexity Cue**: Each recipe card now includes small ingredient icons and a numerical count of ingredients. This acts as a quick heuristic, allowing users to estimate the recipe's complexity (more ingredients often imply more steps and cooking time) at a glance.

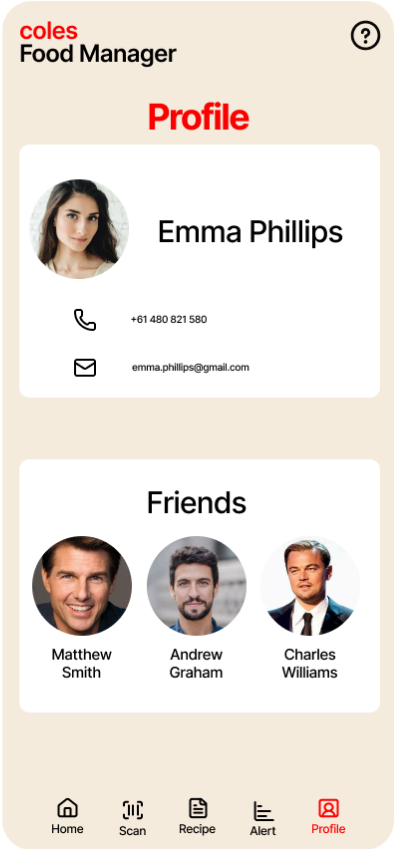
- **Algorithmic Realism**: The generic, visually repetitive "Lemon Tart" placeholder images were replaced with diverse recipes (BBQ Chicken, Tomato Soup, etc). This change avoids visual fatigue and better demonstrates the real-world algorithmic personalization based on the user's inventory.

Complete 'Profile' page development

The Profile page layout was fully developed to effectively display key user information. This provides a clear sense of ownership and identity within the application by featuring:

- A dedicated section for the user's personal contact information (photo, name, phone, email).
- A clear and visually engaging "Friends" section with recognizable photos to display connected family members and friends.





General User Interface Refinements

In addition to the specific changes above, we made several global improvements to optimize the interface:

- **Optimized Page Titles**: Page titles were refined for clarity, conciseness, and aesthetic appeal. For example, changing the title from "Recipe Recommendations" to "Recipe Ideas" is more concise and user-friendly.

- **Aesthetic and Usability Refinement**: We adjusted the alignment, arrangement, and styling of UI components for enhanced usability and aesthetic balance. This involved: reducing the appearance of dense text by adopting a more balanced spacing structure; increasing padding and margin to improve overall visual clarity and readability; adjusting font size, letter spacing and line height of text; making important elements more prominent and scannable.

- Finally, navigation buttons and links were implemented to ensure the interface is interactive and ready for the upcoming usability testing session.

Hi-fi model two (final model)

Conclusion

Throughout our development and designing of our app our team has been able to develop a very strong understanding of human computer interaction principles (HCI). We have also gained practical experience of how to use HCI to address real world problems and in our case for reducing food waste.

Overall achievements

We began with thorough problem identification and PACT analysis, followed by data collection and thematic analysis that grounded our design in real user requirements. We start by establishing paper prototypes to low-fidelity and high-fidelity interactive wireframes. Each iteration was carefully designed by cognitive walkthroughs, usability testing and Think-Aloud evaluations. Through these processes, we implemented intuitive features such as navigation bar, barcode camera, visual indicators, AI recipe model and inventory management. In the end, we were pleased with our demos for our app and showcasing our poster; hearing positive feedback on such a solution.

Challenges and Limitations

The successful implementation of our core sustainability goal, reducing food waste, depends on addressing several key project challenges:

- Mitigating User Testing Bias: Primary reliance on university students for usability and Think-Aloud testing risks biased feedback due to their typically simpler inventory needs. This demographic limitation may have obscured complex feature requirements necessary for multi-user inventory management or high-volume household tracking, adding complexity to the final design.
- Addressing Privacy and Trust: The necessity to track highly personal user data (inventory, consumption, and financial habits) presents a critical design challenge. The final solution must achieve an intricate balance: ensuring data collection is minimal and essential while architecting a transparent, trustworthy user experience that encourages adoption without compromising user control or confidentiality.

For future work we could extend our prototype into a functional app that can be deployed. This would incorporate AI to generate recipes, and generate information on food spoilage. It could also be improved by analysing users consumption patterns and expanding over multiple household members. We could also have further usability testing with a broader demographic so that the application is more accessible. These key improvements and future work will mean our app is more capable of reducing food waste for users at a household level.